The Media Connectivity Association



OVERVIEW AND UPDATES · BROADCAST AND MULTICAST

Public Warning Systems

Emergency warnings delivered over 5G Broadcast

3GPP ETSI

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About the project

Public warning messages over 5G Broadcast: the documents the project rests on

ETSI TS 103 720

Technical Specification

The 5G Broadcast system: it defines support for public warning messages. They are sent via the Commercial Mobile Alert Service, with SIB12 carrying the notification.

3GPP TS 23.041

Technical Specification

The Cell Broadcast Service: message structure, message identifiers, serial numbers and data coding, for ETWS and CMAS.

3GPP TS 36.331

Technical Specification

The warning-message system information blocks and the UE actions on receiving them.

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The CBS message

A warning is a short, self-contained data structure rather than an IP flow, so it reaches idle and non-attached devices

CBS fields populated by the transmitter

- Message Identifier
- Serial Number
- Data Coding Scheme
- Warning Message Segment

SIB12 a CMAS notification

- CMAS alert

SIB10 and SIB11 ETWS primary and secondary notifications

- ETWS notification
- ETWS body

Default alert in the transmitter configuration

- Message identifier 1102 (hex)
- ETWS earthquake and tsunami warning

The Serial Number carries the geographical scope, the message code and the update number: a changed update number signals an updated message for the same event. For PWS, networks may only use identifiers between 4352 and 6399.

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Transmit and receive roles

How the Reference Tools map to the standard on the 5G Broadcast transmit chain

Transmitter

The E-UTRAN cell

Builds the system information on the broadcast carrier and schedules the warning-message SIBs on the radio frame. It plays the role of the eNB in a live network.

SIB12 encoding

From the CBS fields

Builds the SIB12 body from the message identifier, serial number, data coding scheme and warning-message segment. The first implementation uses fixed values for a default alert.

Device

A 5G Broadcast receiver

Decodes the system information, detects the warning-message SIB and presents the alert. There is no alert-specific receiver: a warning is another scheduled SIB.

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How it grew

What each 3GPP release added to public warning, from the release descriptions

Release 15

The 5G System supports public warning through the CBC Function and the AMF. The NG interface gets a warning message transmission function.

Release 16

ePWS adds message identifiers for ePWS-UEs and language-independent content. ETWS and CMAS notification in connected mode for non-Cat-M UEs. TR 36.976 is produced.

Release 17

Public warning is supported over standalone non-public networks (SNPN).

Release 19

PWS over satellite E-UTRAN and NG-RAN, ETWS geo-fencing and duplicate detection with a shared MCC. NB-IoT system information blocks for ETWS and CMAS.

Release 20

Work plan of 28 September 2026: stage 1 for multilingual template-based alerts (MLT-REQ), and CPAS with CPAS-CT in TS 23.041. Unverified: no Release 20 release description.

From the 3GPP release descriptions TR 21.915, TR 21.916, TR 21.917 and TR 21.919; the full list is on the Standards Evolution page.

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Repositories

The alert path, from the Cell Broadcast Centre to the device, from the descriptions of the repositories

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-mbms-tx-for-qrd-and-crd	the 5G Broadcast transmitter, built on srsRAN 4G, for QRD and CRD devices	Transmitter MBMS system information QRD and CRD devices	ea-1.0.0 5 Sep 2024	GitHub
rt-mbms-tx	the standalone LTE-based 5G Broadcast transmitter; shared with TV and Radio	Transmitter MBMS system information	v1.0.0 21 Jun 2023	GitHub
rt-pws-cbc	the Cell Broadcast Centre: takes a warning as JSON, writes a CAP record, sends the warning requests to the MME	Warning input as JSON CAP record Requests to the MME	none yet	GitHub
rt-mbms-application	a receive-side web page that shows the alerts a modem decodes, with their history	CMAS and ETWS alerts Alert history	none yet	GitHub

FIND OUT MORE Technology Specifications Reference Tools

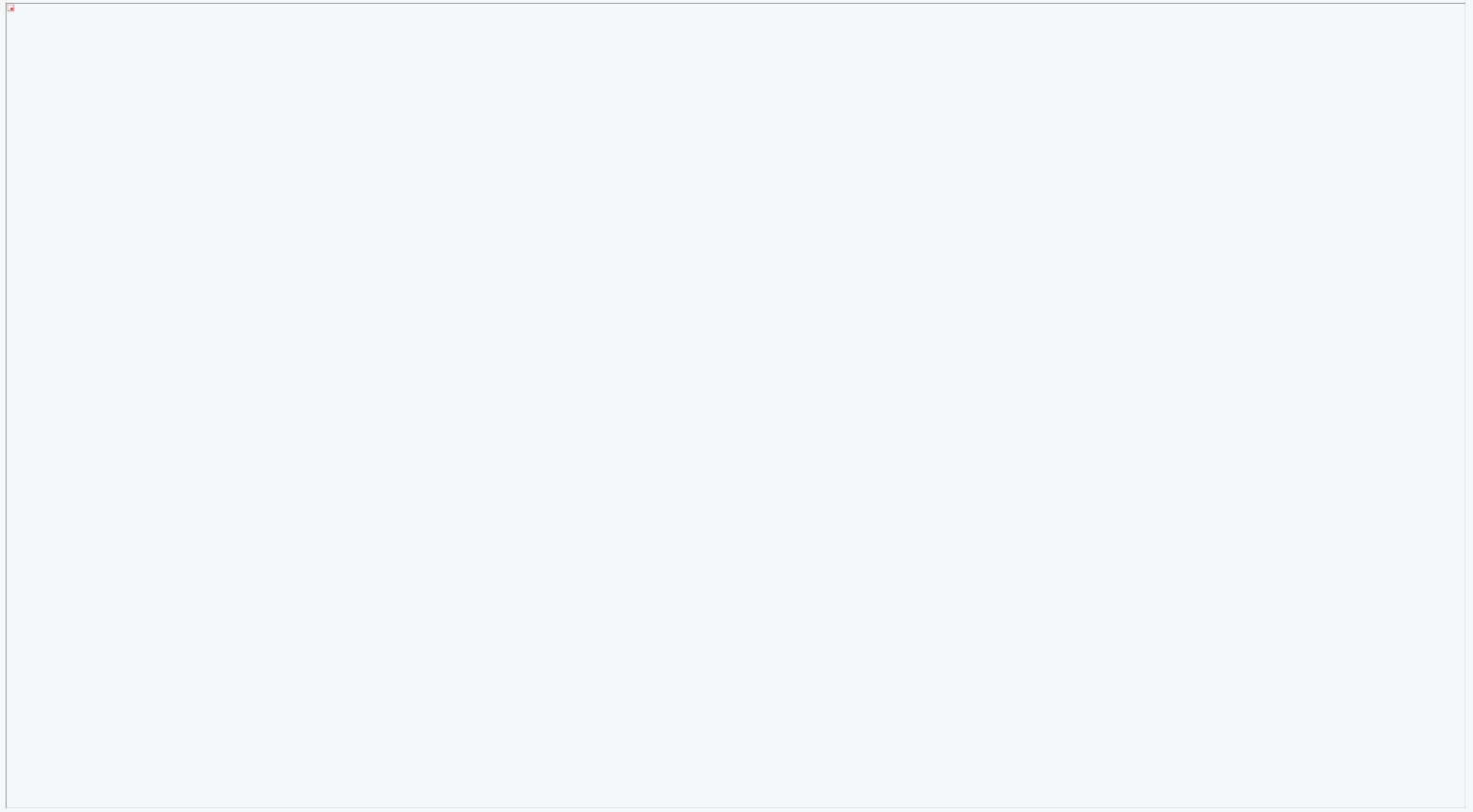
● released ● no release yet

More information at 5g-mag.com/tech/emergency-alerts

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1. **rt-pws-cbc**
2. **rt-mbms-application**
3. **rt-mbms-modem**
4. **rt-mbms-tx-for-qrd-and-crd**

Boxes without a badge are not mapped to a repository. The MME has no repository.

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See it and try it

A bladeRF SDR transmitting, and an earthquake and tsunami warning on a phone



Warning received from the SDR transmitter

CONTRIBUTORS

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WHERE WE STAND

5G Broadcast for Emergency Alerts

SOS

BROUGHT TO STUDY

- Support of 3GPP Service URL to wake up 5G Broadcast receiver on emergency alert reception

ON-GOING

- Cell Broadcast Centre
- Docker support
- Custom alert message and warning type
- ETWS support (SIB10, SIB11)

OUTCOMES

- Emergency alert broadcast
- First release of the LTE-based 5G Broadcast transmitter

RELATION WITH SDOS AND INDUSTRY FORA 3GPP ETSI

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